Thesis Check Meeting2

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Research Focus

- Predicting the scale and trends of social movements.
- Analyze dynamic features of social networks:
  - Information diffusion paths.
  - Emergence of key nodes.
  - Evolution of dynamic structures.
- Objectives:
  - Identify how “critical groups” form.
  - Examine high-impact nodes in the network.
  - Evaluate timing patterns of information diffusion.

Progress Update: Question Definition

- **Core Research Question:**
  - How can dynamic features of social networks predict information diffusion and the spread of social movements?
- No significant changes to the research focus.
- **Key Elements:**
  - Nodes, connections, and their daily changes within the network.

Progress Update: Methods

- **Current Approach:**
  - Collect and scrape Twitter data.
  - Annotate data with sentiment labels.
  - Generate graph data:
    - Retweeters, user relationships, and sentiment features.
  - Analyze large-scale graph datasets using neural networks.
- **Key Goal:**
  - Predict outbreak potential and diffusion trends using early-stage network graphs (e.g., 2 to 4 month).
- **Critical Factors:**
  - High-impact nodes.
  - Emotional intensity.
  - Diversity of diffusion paths.
- **Planned Improvement:**
  - Conduct unsupervised neural network tests to evaluate factor importance.

Progress Update: Data

- **Good News:**
  - Main program running smoothly.
  - Collected 100,000 tweets (May 2020 onward) using Apify.
  - Cost-effective: 10% of official platform cost.
- **Data Details:**
  - Keyword: "Black Lives Matter."
  - Retweet more than 5.
  - Fields: Publisher ID, tweet ID, content, retweets, likes, clicks.

Main Program Data


```
[
    {
        "url": "https://x.com/ramonashelburne/status/1277753534818119680",
        "twitterUrl": "https://twitter.com/ramonashelburne/status/1277753534818119680",
        "id": "1277753534818119680",
        "text": "Story with [U+2066]@ZachLowe_NBA[U+2069] on the NBA and NBPA’s plan to paint 'Black Lives Matter' on courts in Orlando https://t.co/HumecbtEmb
        "retweetCount": 40,
        "replyCount": 30,
        "likeCount": 154,
        "quoteCount": 11,
        "createdAt": "Mon Jun 29 23:59:16 +0000 2020",
        "bookmarkCount": 3,
        "isRetweet": false,
        "isQuote": false
    },
    {
        "url": "https://x.com/ForAmerica/status/1277752808561872896",
        "twitterUrl": "https://twitter.com/ForAmerica/status/1277752808561872896",
        "id": "1277752808561872896",
        "text": "Black Lives Matter is a Marxist organization. \n\nTwitter’s bio supports Marxism.\n\nMarxism is communism. \n\nThank you for attending this TE
        "retweetCount": 26,
        "replyCount": 3,
        "likeCount": 61,
        "quoteCount": 0,
        "createdAt": "Mon Jun 29 23:56:23 +0000 2020",
        "bookmarkCount": 1,
        "isRetweet": false,
        "isQuote": false
    },
    {
        "url": "https://x.com/blackartivist/status/1277752782049640449",
        "twitterUrl": "https://twitter.com/blackartivist/status/1277752782049640449",
        "id": "1277752782049640449",
        "text": "Check out this article from Des Moines Register:\n\nBlack Lives Matter demonstrators rally in Iowa Capitol, push for felon voting rights\n\nht
        "retweetCount": 13,
        "replyCount": 1,
        "likeCount": 38,
```


Figure: Main Program Data

Progress Update: Data

- **Issue:**
  - Testing service for retweeter IDs unavailable (code leak).
  - Developer unresponsive; alternative providers lack full functionality.
- **Potential Solution:**
  - Use Twitter's official API (costly).
  - May reduce data scale within budget constraints.


```
[
    {
        "code": 0,
        "msg": "Due to the leak of apify source code, we have temporarily terminated the service,  Please contact us if you have questions. DC: discord.gg/VEKBj3fT9G tel: https://t.
        "data": "Due to the leak of apify source code, we have temporarily terminated the service,  Please contact us if you have questions. DC: discord.gg/VEKBj3fT9G tel: https://t
    }
]
```

Progress Update: Data

- **Data Annotation:**
  - Label emotional intensity of tweets.
  - Categorize by media format: Text, images, videos.
- **Theoretical Basis:**
  - Emotional Contagion Theory: Emotions spread across networks.
  - Media Richness Theory: Richer formats convey more emotional cues.
- **Method:**
  - Use ChatGPT for initial annotation.
  - Validate results with 1,000–3,000 manually labeled tweets.
- **Advantages:**
  - ChatGPT outperforms many sentiment models.
  - Customized prompts reduce workload and resources.

Feedback 1: How Do External Events Drive Social Movements?

- **External Events:**
  - Short-term, sudden incidents capturing public attention.
  - Example: George Floyd's death in May 2020 sparked protests and global focus.
  - Act as catalysts for societal emotions and grievances.
- **Social Movements:**
  - Long-term, sustained actions with shared goals (e.g., systemic change).
  - Example: "Black Lives Matter" evolved into a global movement.
- **Link:**
  - External events provoke public outrage and empathy.
  - Mobilize individuals for larger, sustained actions (e.g., protests, online advocacy).

Feedback 2: What Is the Dynamic Feedback Mechanism in Network Structures?

- **Definition:**
  - Social media connections (e.g., follows, retweets) evolve over time.
  - Feedback is bidirectional: interactions circulate, amplify, or diminish.
- **Example:**
  - A popular tweet becomes an "issue hub."
  - More users follow the author or adopt the hashtags.
  - Network structure changes (e.g., tighter groups, merged communities).
- **Dynamic Nature:**
  - Repeated interactions create a snowball effect.
  - Nodes and community clusters evolve continuously.
- **Impact:**
  - Drives information diffusion and shapes collective behavior.
  - Supported by Social Network Analysis and complex systems research.

Feedback 3: Pre-Event Network Construction

- **Using Pre-Event Data:**
  - Collect tweets and interactions (e.g., retweets, replies, follows) related to BlackLivesMatter.
  - Focus on early discussions from April 1, 2020, to May 24, 2020 (pre-event phase).
- **Constructing Baseline Network:**
  - Use interactions during the pre-event phase to build the baseline network structure.
  - Capture key metrics: discussion intensity, community distribution, and issue centrality.
- **Defining the Start of "Black Lives Matter":**
  - Start with early attention before May 2020.
  - Compare pre-event and post-event networks to identify shifts and trends.

Feedback 4: Isolating the Impact of the Confounding Event

- **Filtering Confounding Factors:**
  - Exclude content related to mainstream events (e.g., pandemic, elections).
  - Reduce interference from unrelated topics.
- **Identifying Additional Influences:**
  - Use text analysis methods (e.g., frequent itemsets) to detect potential influencing factors.
- **Focusing on the Target Event:**
  - Filter out variables unrelated to the target event.
  - Isolate the event's substantive impact on network structure.

Feedback 5: Sentiment Analysis and Annotation

- **Theoretical Basis:**
  - Emotional Contagion Theory:
    - Emotions categorized as positive, negative, or neutral.
    - Consider emotional intensity for varying diffusion impacts.
  - Media Richness Theory:
    - Text, images, and videos differ in information capacity and emotional expression.
    - Record and analyze formats separately.
- **Annotation Process:**
  - Use ChatGPT for initial annotation.
  - Manually validate 1,000–3,000 tweets to ensure accuracy and consistency.

Feedback 6: Prediction Objectives and Metrics

- **Core Objective:**
  - Simulate the diffusion of topics or emotions using a dynamic network model.
  - Compare predictions with real data to verify accuracy.
- **Key Metrics:**
  - **Nodes and Edges:**
    - Reflect changes in network size and connectivity.
    - Compare predicted node/edge counts with real data on user participation.
  - **Emotional Intensity Distribution:**
    - Measure the spread and impact of emotions within the network.
  - **Diffusion Patterns:**
    - Capture information flow paths and coverage within the network.
- **Validation:**
  - Assess model accuracy by comparing predicted and real metrics over specific time windows.

Feedback 6: Prediction Objectives and Metrics

- **Emotional Intensity:**
  - Predict proportions and intensity changes of positive, negative, and neutral emotions.
  - Compare predictions to actual tweet data using:
    - Mean Squared Error (MSE).
    - Cross-Entropy for distribution alignment.
- **Diffusion Patterns:**
  - Simulate information pathways and coverage within the network.
  - Compare predicted and actual interaction patterns (e.g., retweet paths).
  - Metrics: Jaccard similarity, propagation coverage rates.
- **Validation Metrics:**
  - Assess model performance with:
    - Mean Squared Error (MSE).
    - Mean Absolute Error (MAE).
    - R^2 for network size and emotional diffusion.

Thank You!